

The Bundle-Projection Factorization Converse

Answered in the published version of arXiv:1212.2707

Automated Mathematics Run `fa_banach_001`, lane 0

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Status: literature already answered (affirmative). No new theorem is claimed. The arXiv v2 text asks for the converse of its factorization criterion. Ji and Sarkar's published Theorem 7.4 proves the equivalence after correcting the range from $V(w)$ to $V(\bar{w})$, which is the range of the analytic frame that factors the projection onto $V(\bar{w})$.

1 The arXiv-version question

Let $\mathcal{M}$ be a Cowen–Douglas atom and let $\mathcal{H} \in B_m(\mathbb{D})$ be a pure $\mathcal{M}$ -contractive Hilbert module whose eigenvector bundle has the form

$$E_{\mathcal{H}} \cong E_{\mathcal{M}} \otimes V,$$

where V is an anti-holomorphic rank- m coefficient bundle in a Hilbert space $\mathcal{E}$. Write $\bar{\Pi}_V(w)$ for the orthogonal projection onto $V(\bar{w})$. The arXiv v2 text proves that a factorization

$$\bar{\Pi}_V(w) = F(w)G(w)$$

with bounded F and G implies $\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m$, then asks whether similarity implies such a factorization. Its displayed range condition is $\text{ran } F(w) = V(w)$, although the projection has range $V(\bar{w})$.

Douglas atom does not cover the Dirichlet module (see [20]).

Some of the results of this paper can be generalized in the several variables set up. However, one of the key ideas to achieve results of full strength is closely related to the corona problem in several variables (see [21]).

Another interesting question relates the quasi-affinity of the Cowen-Douglas Hilbert modules. For the Hardy space, quasi-affinity to a submodule of a Hardy module is as same as the Hardy module it self. It is not known under what additional condition on the frame, that module will be quasi-affine to a Cowen-Douglas atom of finite multiplicity.

We also do not known whether the converse of Theorem 7.4 is true. More precisely, let $\mathcal{H} \in B_m(\mathbb{D})$ be a pure $\mathcal{M}$ -contractive Hilbert module, let $E_{\mathcal{H}} \cong E_{\mathcal{M}} \otimes V$, and let $V(w) \subseteq \mathcal{E}$, $w \in \mathbb{D}$, for some Hilbert space $\mathcal{E}$. Assume moreover that $\bar{\Pi}_V(w)$ is the orthogonal projection of $\mathcal{E}$ onto $V(\bar{w})$. Let

$$\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m.$$

Does there exist a Hilbert space $\mathcal{E}_*$, a bounded analytic function $F \in H_{B(\mathcal{E}_*, \mathcal{E})}^\infty(\mathbb{D})$, and a bounded function $G \in L_{B(\mathcal{E}, \mathcal{E}_*)}^\infty(\mathbb{D})$ such that

$$\text{ran } F(w) = V(w),$$

and

$$\bar{\Pi}_V(w) = F(w)G(w),$$

for all $w \in \mathbb{D}$?

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Figure 1: Ji–Sarkar, arXiv:1212.2707 v2, PDF page 18.

2 The published answer

The final journal version replaces the one-way theorem by an if-and-only-if statement and uses the consistent condition

$$\text{ran } F(w) = V(\bar{w}).$$

Thus its Theorem 7.4 states that

$$\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m \iff \exists F \in H_{B(\mathcal{E}_*, \mathcal{E})}^\infty(\mathbb{D}), G \in L_{B(\mathcal{E}, \mathcal{E}_*)}^\infty(\mathbb{D}) : \begin{cases} \text{ran } F(w) = V(\bar{w}), \\ \bar{\Pi}_V(w) = F(w)G(w). \end{cases}$$

SIMILARITY OF QUOTIENT HILBERT MODULES IN THE COWEN-DOUGLAS CLASS 17

Let $\mathcal{H} \in B_m(\mathbb{D})$ be a pure $\mathcal{M}$ -contractive Hilbert module such that $E_{\mathcal{H}} \cong E_{\mathcal{M}} \otimes V$ and $V(w) \subseteq \mathcal{E}$, $w \in \mathbb{D}$. Let $\bar{\Pi}_V(w)$ be the orthogonal projection of $\mathcal{E}$ onto $V(\bar{w})$. In the following theorem, we prove that $\mathcal{H}$ is similar to $\mathcal{M} \otimes \mathbb{C}^m$ if and only if $\bar{\Pi}_V(w)$ can factor as

$$\bar{\Pi}_V(w) = F(w)G(w) \quad (w \in \mathbb{D}),$$

for some $F \in H_{B(\mathcal{E}_*, \mathcal{E})}^\infty(\mathbb{D})$ and $G \in L_{B(\mathcal{E}, \mathcal{E}_*)}^\infty(\mathbb{D})$ with $\text{ran } F(w) = V(w)$, $w \in \mathbb{D}$.

THEOREM 7.4. *Let $\mathcal{E}$ and $\mathcal{E}_*$ be two Hilbert spaces, let $\mathcal{H} \in B_m(\mathbb{D})$ be a pure $\mathcal{M}$ -contractive Hilbert module such that*

$$E_{\mathcal{H}} \cong E_{\mathcal{M}} \otimes V,$$

where $V(w) \subseteq \mathcal{E}$. Assume that $\bar{\Pi}_V(w)$ is the orthogonal projection of $\mathcal{E}$ onto $V(\bar{w})$. Then

$$\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m$$

if and only if there exist $F \in H_{B(\mathcal{E}_, \mathcal{E})}^\infty(\mathbb{D})$ and $G \in L_{B(\mathcal{E}, \mathcal{E}_*)}^\infty(\mathbb{D})$ such that*

$$\text{ran } F(w) = V(\bar{w}) \quad (w \in \mathbb{D})$$

and

$$\bar{\Pi}_V(w) = F(w)G(w) \quad (w \in \mathbb{D}).$$

Proof. First we prove that the conditions are sufficient. Note that

$$\text{ran } F(w) = V(\bar{w}) = \text{ran } \bar{\Pi}_V(w),$$

implies that

$$\bar{\Pi}_V(w)F(w) = F(w) \quad (w \in \mathbb{D}).$$

Note also that (cf. (7.1))

$$\partial \bar{\Pi}_V(w)F(w) = (1 - \bar{\Pi}_V(w))F'(w).$$

and (cf. Lemma 2.2, [21])

$$\partial \bar{\Pi}_V(w) = \partial \bar{\Pi}_V(w)\bar{\Pi}_V(w),$$

for all $w \in \mathbb{D}$. We therefore have

$$\begin{aligned} \partial \bar{\Pi}_V(w) &= \partial \bar{\Pi}_V(w)\bar{\Pi}_V(w) \\ &= \partial \bar{\Pi}_V(w)F(w)G(w) \\ &= (1 - \bar{\Pi}_V(w))F'(w)G(w) \end{aligned}$$

Then for $M = \|G\|_\infty > 0$, it follows that

$$\begin{aligned} \|\partial \bar{\Pi}_V(w)\|_2 &= \|(I - \bar{\Pi}_V(w))F'(w)G(w)\|_2 \\ &\leq \|I - \bar{\Pi}_V(w)\| \cdot \|F'_2(w)G(w)\|_2 \\ &\leq \|G(w)\| \cdot \|F'(w)\|_2 \\ &\leq \|G\|_\infty \|F'(w)\|_2 \\ &= M \|F'(w)\|_2, \end{aligned}$$

for all $w \in \mathbb{D}$. Now as in the proof of Theorem 7.2, we set the bounded subharmonic function ψ_1 as $\|M^{\frac{1}{2}}F_2\|_2^2$. Then

$$\text{trace } K_{\mathcal{H}_1} - \text{trace } K_{\mathcal{M} \otimes \mathbb{C}^m} = \|\partial \bar{\Pi}_{V_1}(w)\|_2^2 \leq \Delta \psi_1,$$

Figure 2: Published version, Theorem 7.4, PDF page 17.

3 Why the converse holds

Similarity gives, through the paper's Theorem 7.2, a uniformly bounded and uniformly bounded-below anti-holomorphic bundle frame

$$\Gamma(w) : \mathbb{C}^m \longrightarrow \mathcal{E}, \quad \text{ran } \Gamma(w) = V(w).$$

Set

$$F(w) = \Gamma(\bar{w}), \quad G(w) = (F(w)^* F(w))^{-1} F(w)^*.$$

Then F is bounded analytic, the lower frame bound makes G bounded, and the standard projection formula gives

$$F(w)G(w) = F(w)(F(w)^* F(w))^{-1} F(w)^* = P_{\text{ran } F(w)} = P_{V(\bar{w})} = \bar{\Pi}_V(w).$$

This is precisely the requested converse in its type-consistent form.

18

KUI JI AND JAYDEB SARKAR

and hence by (iii) $\implies$ (i) of Theorem 7.2, we have that

$$\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m.$$

Conversely, assume that $\mathcal{H} \sim_s \mathcal{M} \otimes \mathbb{C}^m$. By (i) $\implies$ (ii) of Theorem 7.2, there exist an anti-holomorphic point-wise invertible bundle map $\Phi : E_{\mathcal{M} \otimes \mathbb{C}^m} \rightarrow E_{\mathcal{H}}$ and a one-one bounded anti-holomorphic map $\Gamma : \mathbb{D} \rightarrow \mathcal{B}(\mathbb{C}^m, \mathcal{E})$ such that

$$\Phi(w)(k_{\mathcal{M}}(\cdot, w) \otimes \eta) = k_{\mathcal{M}}(\cdot, w) \otimes \Gamma(w)\eta,$$

for all $\eta \in \mathbb{C}^m$ and $w \in \mathbb{D}$. We have therefore

$$\Phi^{-1}(w)(k_{\mathcal{M}}(\cdot, w) \otimes \Gamma(w)\eta) = k_{\mathcal{M}}(\cdot, w) \otimes \eta,$$

and

$$V(w) = \text{ran } \Gamma(w),$$

for all $\eta \in \mathbb{C}^m$ and $w \in \mathbb{D}$. Let

$$F(w) = \Gamma(\bar{w}),$$

and

$$G(w) = (F^*(w)F(w))^{-1}F^*(w),$$

for all $w \in \mathbb{D}$. Then by (7.2), there exists $\delta > 0$ such that

$$\delta^{-1} \leq \|F\| \leq \delta,$$

and

$$\|G\| \leq \delta^3.$$

Since Γ is anti-holomorphic, it follows that $F \in H_{B(\mathbb{C}^m, \mathcal{E})}^\infty(\mathbb{D})$ and $G \in L_{B(\mathcal{E}, \mathbb{C}^m)}^\infty(\mathbb{D})$. Moreover

$$\begin{aligned} \bar{\Pi}_V(w) &= F(w)(F^*(w)F(w))^{-1}F^*(w) \\ &= F(w)G(w) \end{aligned}$$

for all $w \in \mathbb{D}$. This completes the proof of the result. $\blacksquare$

8. CONCLUDING REMARKS

A number of questions and directions remain to be explored, including the similarity problem for the Dirichlet module (however, see [15]). We point out that the notion of the Cowen-Douglas atom does not cover the Dirichlet module (see [20]).

Some of the results of this paper can be generalized in the several variables set up. However,

Figure 3: Published version, converse proof on PDF page 18.

4 Provenance and scope

The arXiv source and PDF page 18 were checked directly. The authors' final PDF was checked on pages 17–18, and the journal metadata identifies the article as European Journal of Mathematics 5 (2019), 1331–1351, DOI 10.1007/s40879-018-0297-y. This packet records a version-to-publication resolution only. It does not claim a new proof, and it does not address the separate quasi-affinity and several-variable programs in the concluding remarks.

Reference. K. Ji and J. Sarkar, *Similarity of quotient Hilbert modules in the Cowen–Douglas class*, arXiv:1212.2707 v2; European Journal of Mathematics 5 (2019), 1331–1351.